

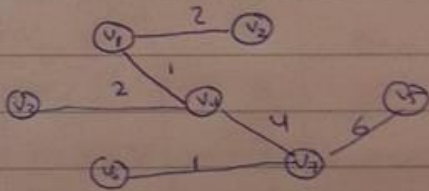
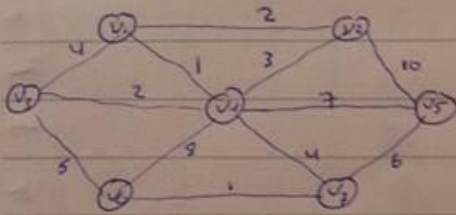
Question Four [20 marks]

Write the best algorithm to solve the following equation:

$$F(x) = x^{12} + 7x^8 + 5x^5 + 12x^2 + 9.$$

أقرب طريقه تروا مكة ومكة

نادر علی قاسم



min cost $\frac{\text{marginal}}{\text{dwr}}$

prim's Algorithm

min unknown distance

Known	d	Pv	k	d	p
v ₁					
v ₂					
v ₃					
v ₄					
v ₅					
v ₆					
v ₇					

MIDTERM

one dimension
array

Knapsack alg ①

② "نسبة القيمة بكمية"
 "أقصى مبلغ - المال وزعمه بأفضل طريقة"

ex 50 50 50

5

1

greedy algorithm

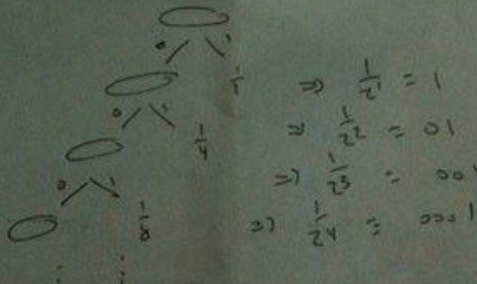
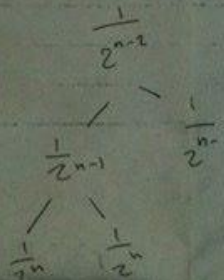
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والتي بعده

...

term 2 $\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{16}$ $\frac{1}{32}$
 "أكتب شجرة قبل ذلك"

... $\frac{1}{2^{n-1}}$ $\frac{1}{2^{n-1}}$
 "أول شجرة قبل ذلك"



$$\frac{1}{2^n} \rightarrow \frac{1000...1}{2^{n-1}}$$

-
- This question paper is composed of 5 pages containing five questions.
 - Please read the whole paper before starting to answer.
 - You have ten minutes to read the paper and then only twenty minutes to ask questions.
 - The only questions allowed concern the spelling and the meaning of English words.
 - Intermediate reasoning steps are demanded. Any unjustified answer is not accepted.
 - Remember to write down you're your full name and your student id number.
-

Question One [20 marks]

Assume you are given a number X and two sorted lists of n number each.
 $A_1 \leq A_2 \leq \dots \leq A_n$ and $B_1 \leq B_2 \leq \dots \leq B_n$. Design a linear (i.e. $O(n)$) time algorithm to determine how many pairs (i, j) there are such that $A_i + B_j \leq X$. Justify the correctness and running time of your algorithm.

Question Three [20 marks]

We are given 5 matrices A, B, C, D, and E with the following dimensions:

A is a 3×5 matrix

B is a 5×4 matrix

C is a 4×6 matrix

D is a 6×7 matrix

E is a 7×3 matrix

Using the dynamic programming technique we arrived at the following matrix with last entry (1,5).

j \ i	2	3	4	5
1	60/1	132/2	258/3	?
2		120/2	308/2	258/2
3			168/3	198/3
4				126/4

Find the matrix grouping that gives the minimum cost.

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Question One [20 marks]

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Dynamic Programming (Longest Common Subsequence)

LCS-LENGTH(X, Y)

```

1   $m \leftarrow \text{length}[X]$ 
2   $n \leftarrow \text{length}[Y]$ 
3  for  $i \leftarrow 1$  to  $m$ 
4    do  $c[i, 0] \leftarrow 0$ 
5  for  $j \leftarrow 0$  to  $n$ 
6    do  $c[0, j] \leftarrow 0$ 
7  for  $i \leftarrow 1$  to  $m$ 
8    do for  $j \leftarrow 1$  to  $n$ 
9      do if  $x_i = y_j$ 
10         then  $c[i, j] \leftarrow c[i-1, j-1] + 1$ 
11             $b[i, j] \leftarrow \text{"\diagup"}$ 
12         else if  $c[i-1, j] \geq c[i, j-1]$ 
13            then  $c[i, j] \leftarrow c[i-1, j]$ 
14                 $b[i, j] \leftarrow \text{"\uparrow"}$ 
15            else  $c[i, j] \leftarrow c[i, j-1]$ 
16                 $b[i, j] \leftarrow \text{"\leftarrow"}$ 
17  return  $c$  and  $b$ 
  
```

إذا كانت $i-1$ و $j-1$ أكبر من Max + 1

إذا كان $c[i-1, j] \geq c[i, j-1]$ Max من فوقه
 إذا كان $c[i, j-1] > c[i-1, j]$ Max من اليسار

إذا كان $c[i-1, j-1] + 1$ (الذي هو Max)

إذا كان $c[i, j] + 1$ Max من فوقه

j	0	1	2	3	4	5	6
i	y_j	B	D	C	A	B	A
0	x_i	0	0	0	0	0	0
1	A	0	0	0	0	1	1
2	B	0	1	1	1	2	2
3	C	0	1	1	2	2	2
4	B	0	1	1	2	2	3
5	D	0	1	2	2	2	3
6	A	0	1	2	2	3	4
7	B	0	1	2	2	3	4

إذا كان $c[i, j] + 1$ Max من فوقه

PRINT-LCS(b, X, i, j)

```

1  if  $i == 0$  or  $j == 0$  ← فارغة
2  return
3  if  $b[i, j] == \text{"\diagup"}$  ← متساويان
4    PRINT-LCS( $b, X, i-1, j-1$ )
5    print  $x_i$ 
6  elseif  $b[i, j] == \text{"\uparrow"}$ 
7    PRINT-LCS( $b, X, i-1, j$ )
8  else PRINT-LCS( $b, X, i, j-1$ )
  
```

(LCS) Length

Row by Row

إذا كان الفوقها مساويها بنقطتها فوقها مع

Dijkstra, graph, vertices, edges

FINAL

① 2 arrays A, B $A_1 \leq A_2 \leq \dots \leq A_n$

$B_1 \leq B_2 \leq \dots \leq B_n$

and a number X , write algorithm to give number of (i, j) that $A_i + B_j \leq X$.

② Prim's algorithm (splaying tree)

Make the table then draw

③ Cost table and Path $\Rightarrow$ Find & write formula

④ Parallel processing architectures

- Meaning of degree of parallelism

- Difference Between Perfect shuffle & Exchange Shuffle

⑤ Polynomial $2x^{12} + 7x^8 + 5x^5 + 2x^2 + 9$

Find Best algorithm to Solve

⑥ Explain, how these number can be represented in Huffman code with this 8 numbers and then general the rule.

1 1 2 3 5 8 13 21

⑦ Find the following spanning tree for this graph

$\Rightarrow$ you need to use the table to solve it

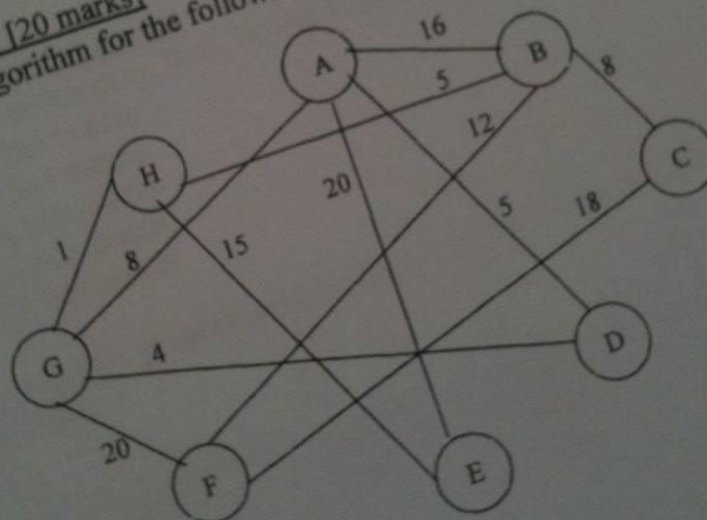
He will give you any graph.

{The prime algorithm and to use}

Question Five [20 marks]

- 1) All computer architectures can be classified into four categories. Describe these categories in details?
- 2) What we mean by **degree of parallelism**?
- 3) Parallel processing topology is the way by which the processors in parallel computers are connected. Describe in details the **Perfect Shuffle and Shuffle Exchange** topology and mention the different between Perfect Shuffle and Shuffle Exchange.

Question Two [20 marks]
Trace prim's algorithm for the following graph. Take vertex A as the starting point.



Finding a missing number

An array of n elements contains all but one of the integers from 1 to $n+1$.

1. Give the best algorithm you can for determining which number is missing if the array is sorted, and analyze its asymptotic worst-case running time.
2. Give the best algorithm you can for determining which number is missing if the array is not sorted, and analyze its asymptotic worst-case running time.

→ Array a;

~~LC~~
$$u \in \mathbb{R}$$
$$\text{mid} = (\text{first} + \text{last}) / 2;$$

divide (A, m, last)

divide(A, ~~m~~, first, m)

divide (A, ~~q~~, n)

divide(A, start, end)

$$1 \rightarrow 11$$

10 3 A 21 5

1	2	3	5	6	7	8	9	10	$\frac{m}{n}$
---	---	---	---	---	---	---	---	----	---------------

قسم ۱ A من النكاح ← ويلحقها كل سنة مستأجرة

$$\boxed{1 \mid 2 \mid 3 \mid 5}$$

6	7	8	9	10
---	---	---	---	----

1 | 2

3/5

6	7
---	---

8/5/16

2

3 5

6

7

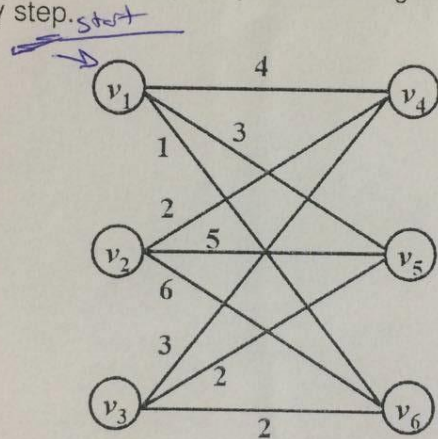


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⑧ 同

Question Four [25 marks]

Use Prim's algorithm to find a minimum spanning tree of the graph in Figure below with v_1 as the root. What is the time complexity of Prim's algorithm? Note that you should present your answer step by step.



	K ₁ A ₁ P ₁	K ₂ A ₂ P ₂	K ₃ A ₃ P ₃	K ₄ A ₄ P ₄	K ₅ A ₅ P ₅	K ₆ A ₆ P ₆
1	0 0 -	1 0 -	1 0 -	1 0 -	1 0 -	1 0 - ✓
2	0 0 -	0 0 -	0 6 V ₆	0 6 V ₆	0 5 V ₅	0 2 V ₄ 1 2 V ₄ ✓
3	0 0 -	0 0 -	0 0 V ₁	1 0 V ₁	0 0 V ₁	0 0 V ₁

Find the value of frequency of character l.

Char	freq.	encoding
B	2	01111
F	1	01110
H	3	0110
I	?	00
L	5	010
M	15	10
S	15	11

الموجود ٦ اذ (هتار ١٥) يسار و اذ ١ (١٥) الحين

① Give an algorithm using dynamic programming to determine how many distinct ways there are to give a cents in change using any coins from among pennies, nickels, dimes, and quarters. For example, there are 6 ways to give 16 cents change:

- A dime, a nickel, and a penny;
- A dime and 6 pennies;
- 3 nickels and a penny;
- 2 nickels and 6 pennies;
- One nickel and 11 Pennies;
- And 16 pennies.

• And 16 pennies.

2 Demonstrate your solution by showing a step-by-step solution for 12 cents change.

first thing, you have to ~~set~~ put the coins in an array.

8) sort them decendingly)

Coins $[] = \{1, 5, 10, 25\}$

then you have to ~~the~~ decide the certs you want to change.

by the coins elements.

then check if you get the number you want

MIDTERM

one dimension
array

Knapsack alg ①

② "نسبة القيمة بكمية"
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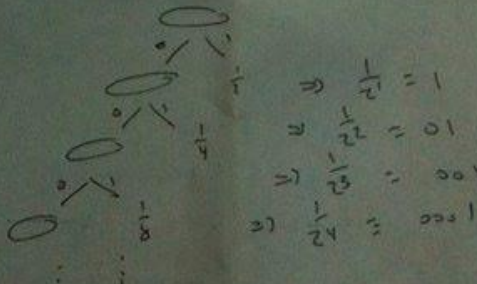
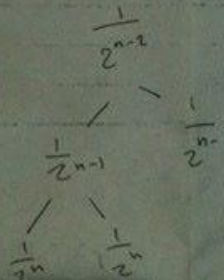
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Algorithm

A : 2 ---> 3 ---> 5 ---> 8 ---> 9 ---> 10

B : 1 ---> 4 ---> 6 ---> 8 ---> 20 ---> 30

Take pointers **i** and **j** so that **i** points to the start of the list A and **j** points to the start of B.
Find the sum $A[i] + B[j]$ and compare it with K
•if $A[i] + B[j] \leq K$ we have found the pair $A[i]$ and $B[j]$

```
i = A.start;
j = B.start;
Var counter = 0;
While(i!=null)
{
    if(Ai + Bj <= X)
    {
        j = j → next;
        counter++;
    }
    else
    {
        i = i → next ;
        j = A.start;
    }
}
```